

# Engineering Principles and Practice I

by Seojin

## Contents

|          |                                     |          |
|----------|-------------------------------------|----------|
| <b>1</b> | <b>Fundamentals of Electricity</b>  | <b>1</b> |
| 1.1      | Electrical quantities . . . . .     | 1        |
| 1.2      | Example circuit . . . . .           | 2        |
| <b>2</b> | <b>Thevenin Equivalent Circuits</b> | <b>2</b> |
| <b>3</b> | <b>Node Voltage Analysis</b>        | <b>2</b> |
| <b>4</b> | <b>Capacitors</b>                   | <b>2</b> |
| <b>5</b> | <b>Capacitors and Inductors</b>     | <b>2</b> |
| <b>6</b> | <b>AC Circuits and Phasors</b>      | <b>3</b> |

## 1 Fundamentals of Electricity

### 1.1 Electrical quantities

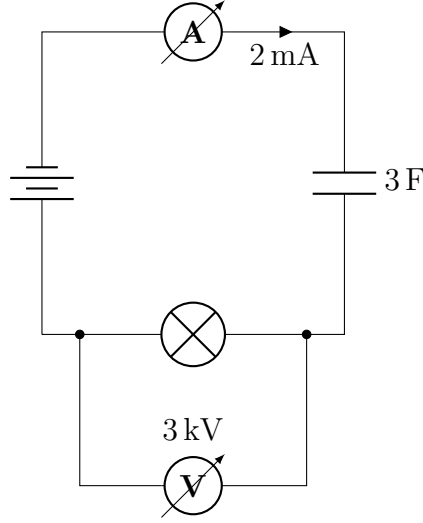
| Quantity   | Description                                          | SI Unit  |
|------------|------------------------------------------------------|----------|
| Charge     | The amount of electricity held by an object          | C        |
| Current    | The rate of flow of charge                           | A        |
| Voltage    | The potential difference between two points          | V        |
| Resistance | The opposition to the flow of current                | $\Omega$ |
| Power      | The rate at which energy is transferred or converted | W        |

$$V = \frac{W}{Q}, \quad I = \frac{dQ}{dt}, \quad P = VI. \quad (1)$$

Ohm's law is

$$V = IR. \quad (2)$$

## 1.2 Example circuit



## 2 Thevenin Equivalent Circuits

Any linear two-terminal network can be replaced by a Thevenin voltage source  $V_{\text{th}}$  in series with a resistance  $R_{\text{th}}$ :

$$V_{\text{th}} = V_{\text{oc}}, \quad R_{\text{th}} = \frac{V_{\text{oc}}}{I_{\text{sc}}}. \quad (3)$$

## 3 Node Voltage Analysis

Choose a reference node, assign a voltage to every remaining node, and apply Kirchhoff's current law:

$$\sum_k \frac{V_n - V_k}{R_k} = I_{\text{injected}}. \quad (4)$$

## 4 Capacitors

For a capacitor,

$$q = Cv, \quad i = C \frac{dv}{dt}, \quad w = \frac{1}{2} Cv^2. \quad (5)$$

For series and parallel combinations,

$$\frac{1}{C_{\text{series}}} = \sum_i \frac{1}{C_i}, \quad C_{\text{parallel}} = \sum_i C_i. \quad (6)$$

## 5 Capacitors and Inductors

An inductor satisfies

$$v = L \frac{di}{dt}, \quad w = \frac{1}{2} Li^2. \quad (7)$$

The impedance of ideal reactive components at angular frequency  $\omega$  is

$$Z_C = \frac{1}{j\omega C}, \quad Z_L = j\omega L. \quad (8)$$

## 6 AC Circuits and Phasors

For sinusoidal steady-state analysis, represent signals using phasors:

$$v(t) = V_m \cos(\omega t + \phi) \quad \longleftrightarrow \quad \tilde{V} = V_{\text{rms}} \angle \phi. \quad (9)$$

Impedances can be combined using the same series and parallel rules as resistances. The average power is

$$P = V_{\text{rms}} I_{\text{rms}} \cos \phi. \quad (10)$$